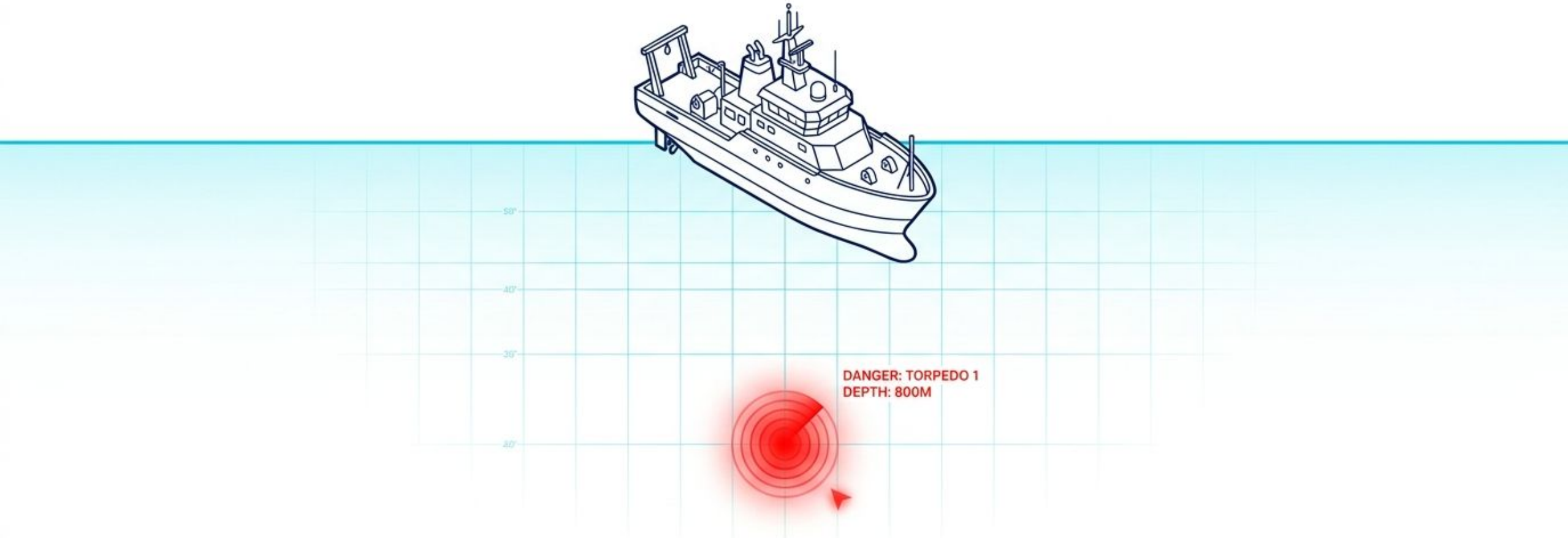
Migration Diagnostic

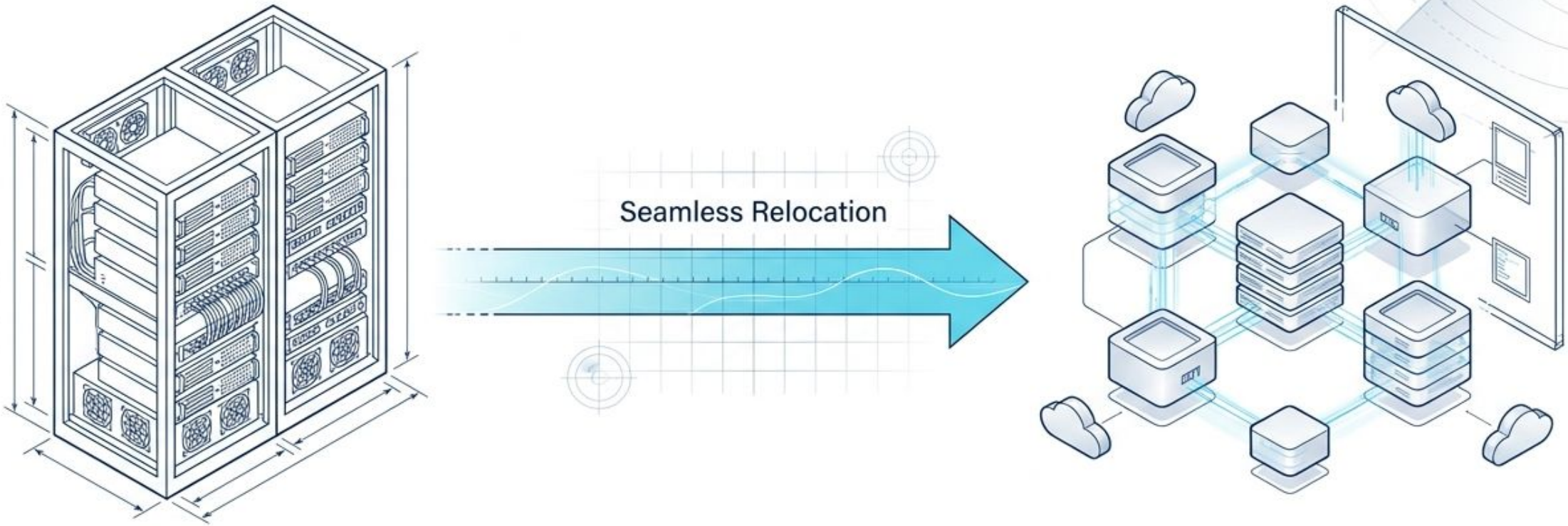
Mitigating the Risk of Migration

Red Hat

Navigating the Depths: Exposing the 6 Hidden Torpedoes of VMware Cloud Migrations



The Lift-and-Shift Illusion: Assuming a Calm Surface



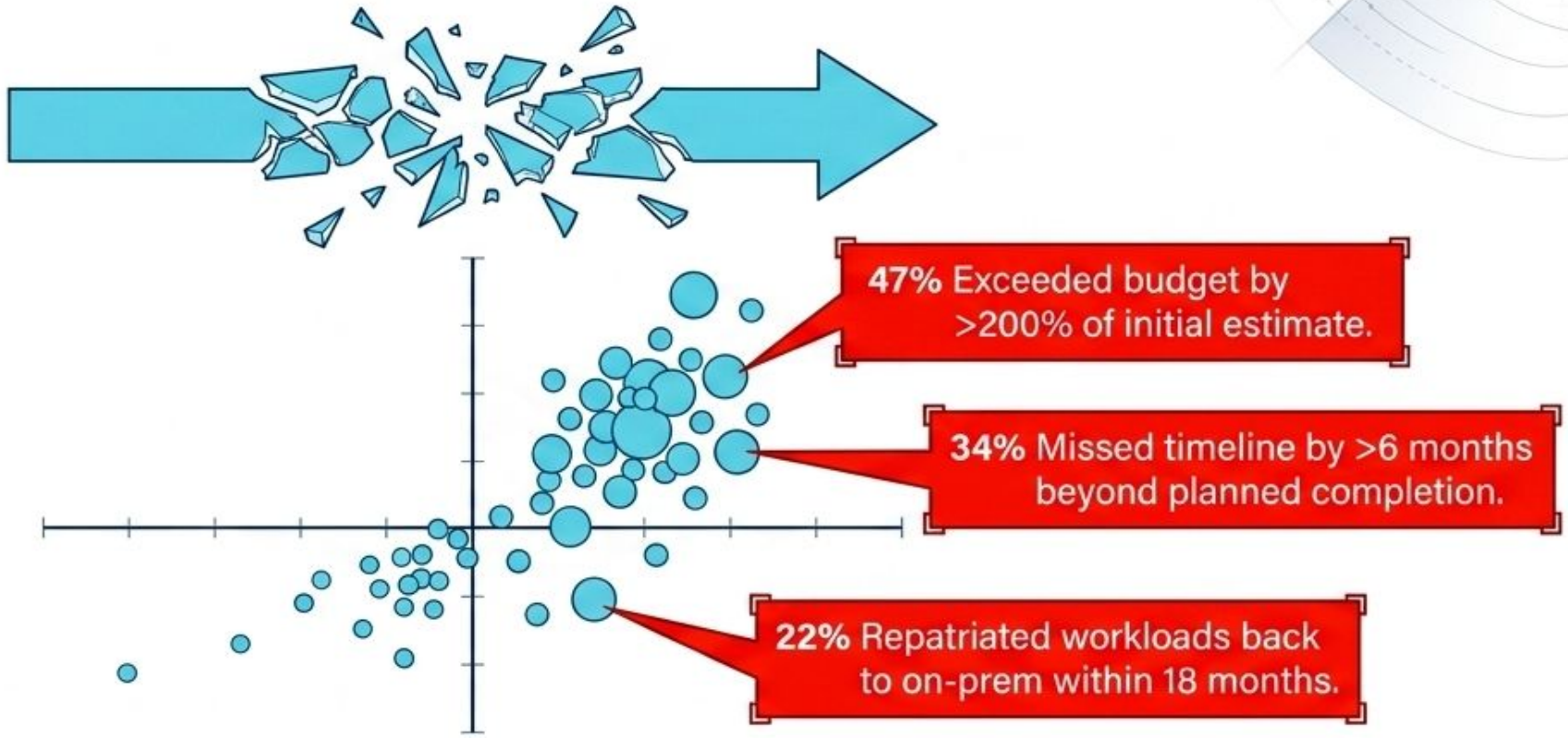
The Convenient Lie

A virtual machine humming along on VMware is the exact same as an EC2 instance.

The False Assumption

We can simply copy-paste our infrastructure from CapEx to OpEx without redesigning our network trust models or addressing accumulated technical debt.

The Ground Truth: 81% of Migrations Fail to Meet Original Objectives



Core Insight: 73% of failed projects shared one root cause—they discovered critical dependencies after the migration launched.

The Lift-and-Shift Myth vs. Ground Truth Diagnostic

The Convenient Lies	The Expensive Truths
The app is completely stateless.	On-prem milliseconds of latency balloon into tens of milliseconds across cloud Availability Zones, breaking implicit trust.
Networking will figure itself out.	Legacy firewall rulesets with thousands of lines cannot translate directly into cloud security groups without triggering massive exposure.
We will fix technical debt after we are in the cloud.	Inefficient on-prem apps using free idle CPU cycles translate directly to spiraling monthly OpEx cloud billing.

The Threat Matrix: Categorizing the Hidden Dependencies

Category 1: Architectural Integrity
(Threats 1 & 4)

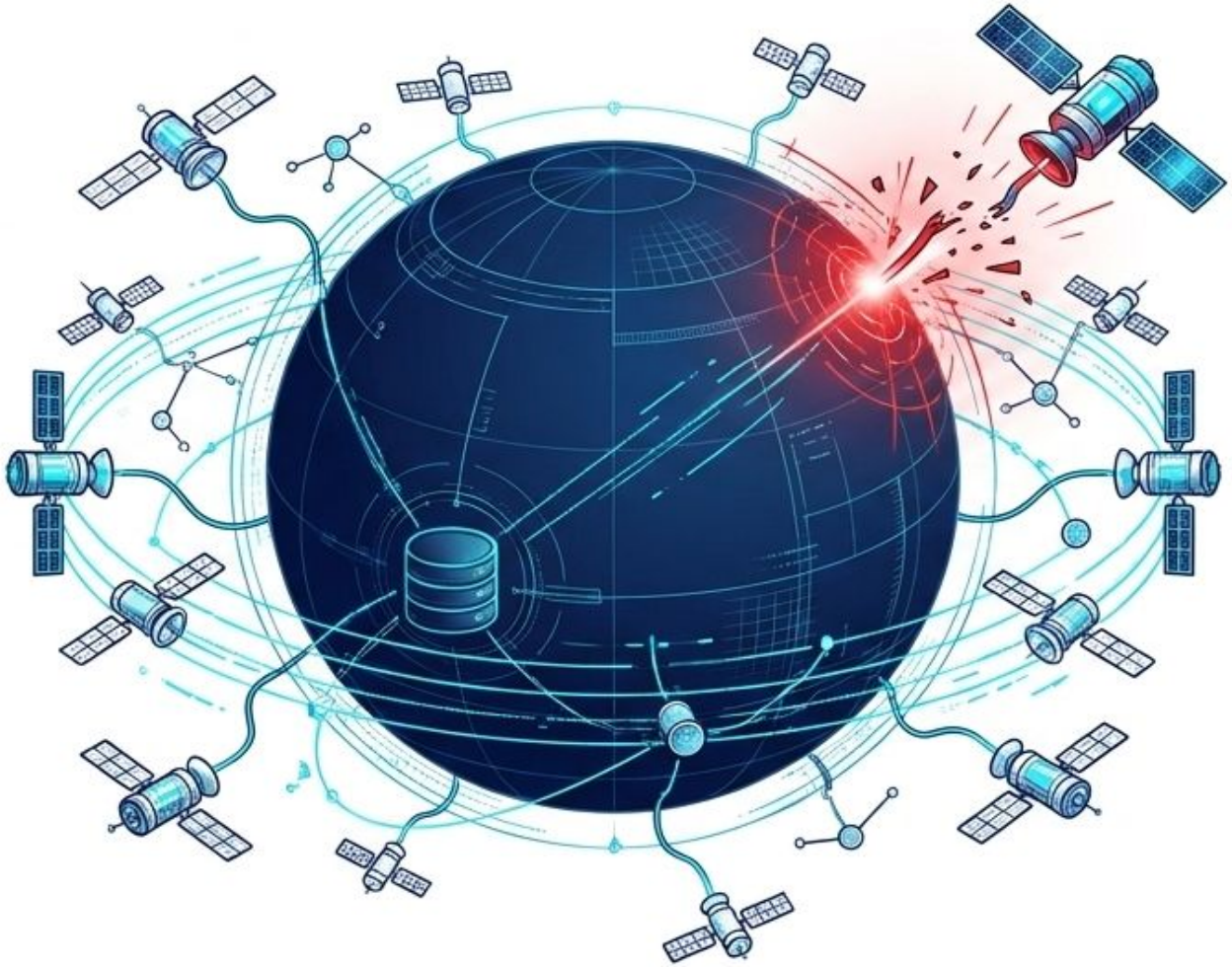
Category 2: Governance & Financial
(Threats 2, 3, & 5)

Category 3: Human & Political
(Threat 6)

These **six dependencies** remain **invisible** to standard automated scans.
They require deliberate Week-Zero sonar to detect.

NotebookLM

Threat 01: Hidden Database Fan-Out and Brittle Read Replicas



The Threat:

You migrate what looks like a single app, only to discover its database is the **secret backbone** for **ten undocumented applications across different VPCs.**

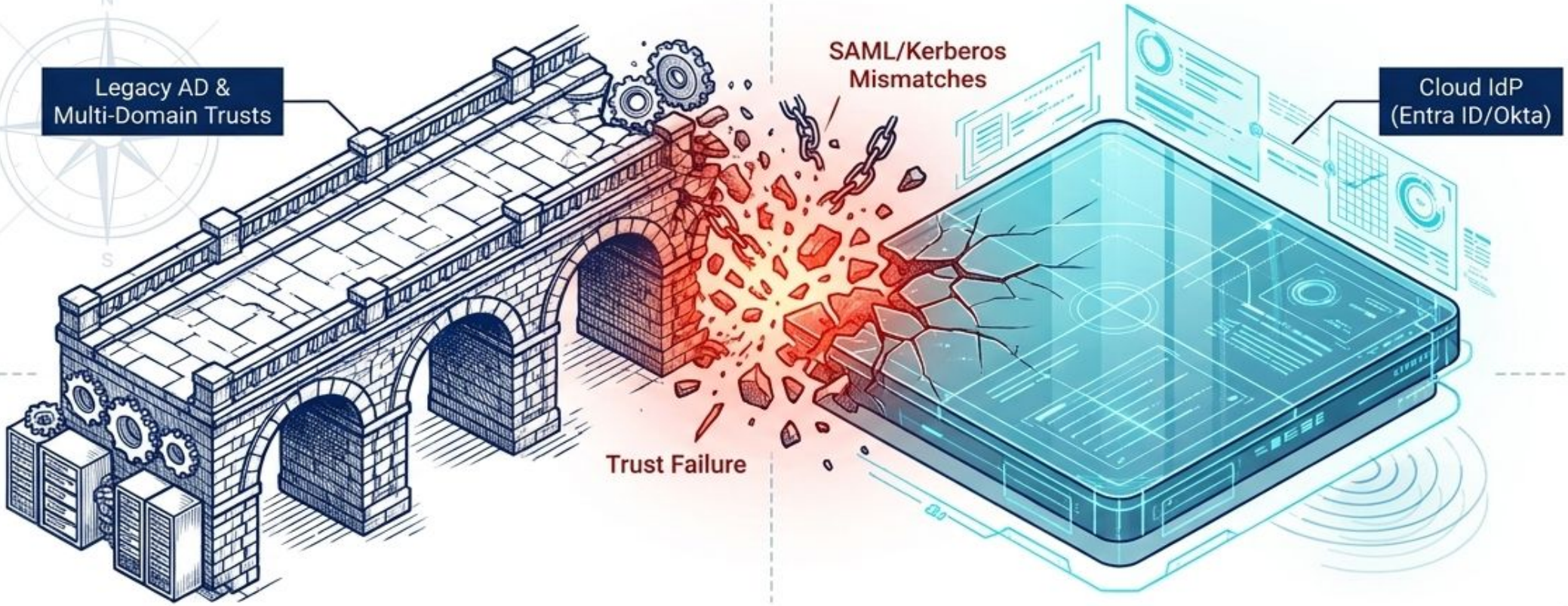
The Blast Radius:

Massive **latency spikes** as **dependent** services crawl across new cloud network boundaries. Performance grinds to a halt.

Sonar Signature:

Look for suspiciously high distinct IP addresses hitting database ports. Watch for off-the-charts **Data Gravity** calculations (TB × ms).

Threat 02: Active Directory & LDAP Trusts That Shatter in the Cloud



The Threat

Older .NET and Java systems rely on complex, multi-domain AD trusts for authentication that have no clean equivalent in cloud-native IdPs.

The Blast Radius

The naïve assumption that we'll just use SAML shatters when you discover the app relies on Kerberos delegation.

Sonar Signature

This single, hidden dependency can instantly halt the migration of an entire legacy application portfolio.

Threat 03: The Financial Toxicity of License-Bound SQL and Windows

The Threat

A perpetual license perfectly happy in your data center may not be portable to a public cloud without triggering massive new consumption-based fees. Bring Your Own License (BYOL) is frequently blocked.

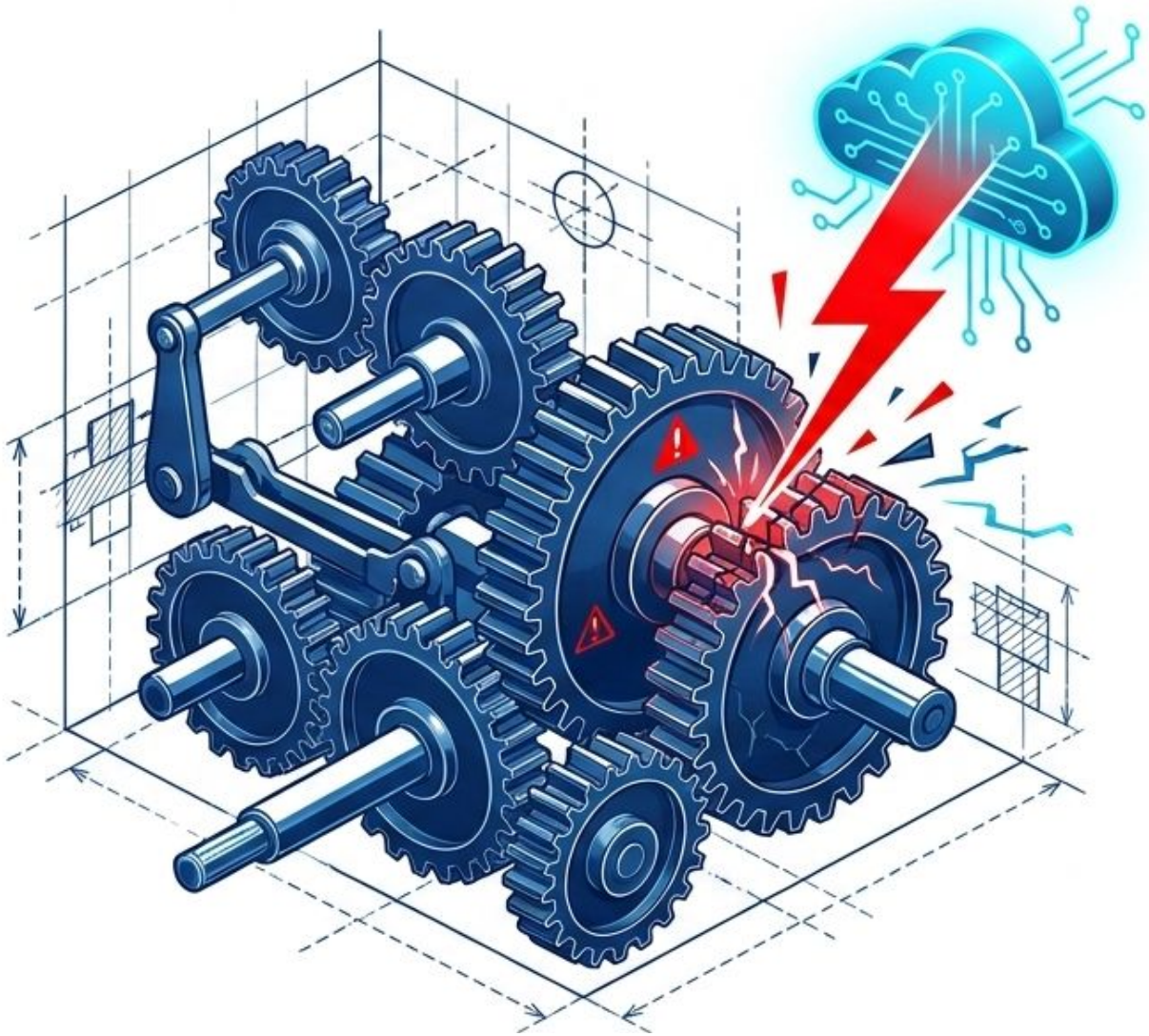
The Blast Radius

Total Cost of Ownership (TCO) triples overnight. The new SQL Server license costs more than the entire infrastructure bill.

The 3x Rule

If modeled license costs are more than 3x the infrastructure costs, rehosting is mathematically the wrong move.

Threat 04: Mainframe Batch Schedules and the 4 A.M. Myth



The Threat

Mainframe apps operate on rigid batch jobs optimized over decades, assuming exclusive, bare-metal access to resources at specific times (e.g., 4 a.m. EST).

The Blast Radius

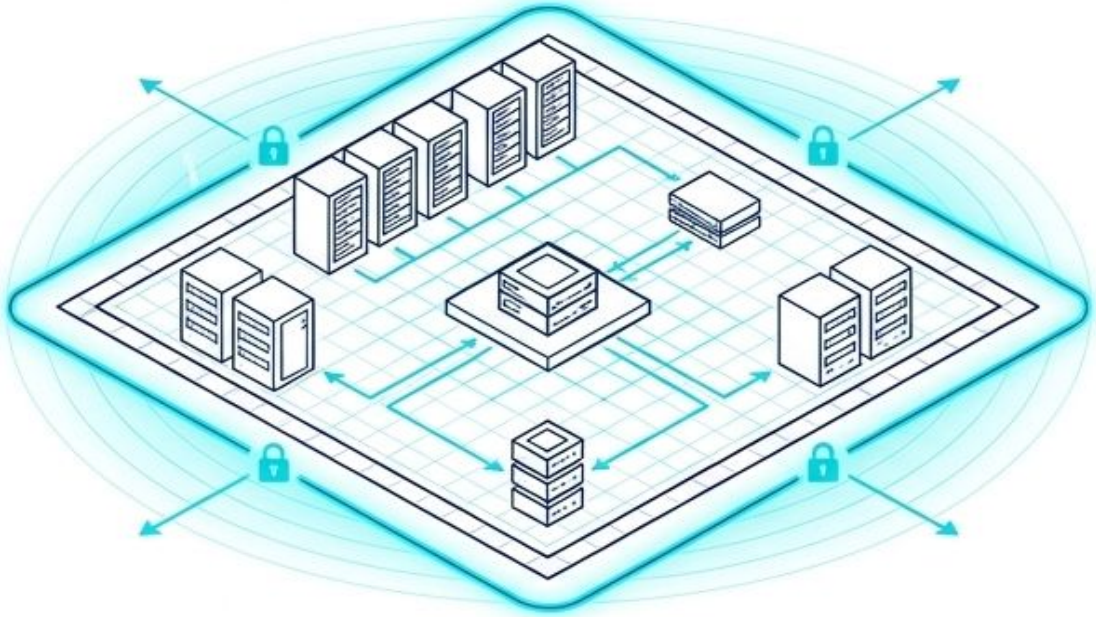
Distributed cloud architectures introduce network latency and jitter. Batch windows are missed, leading to cascading data corruption and business process failures.

Sonar Signature

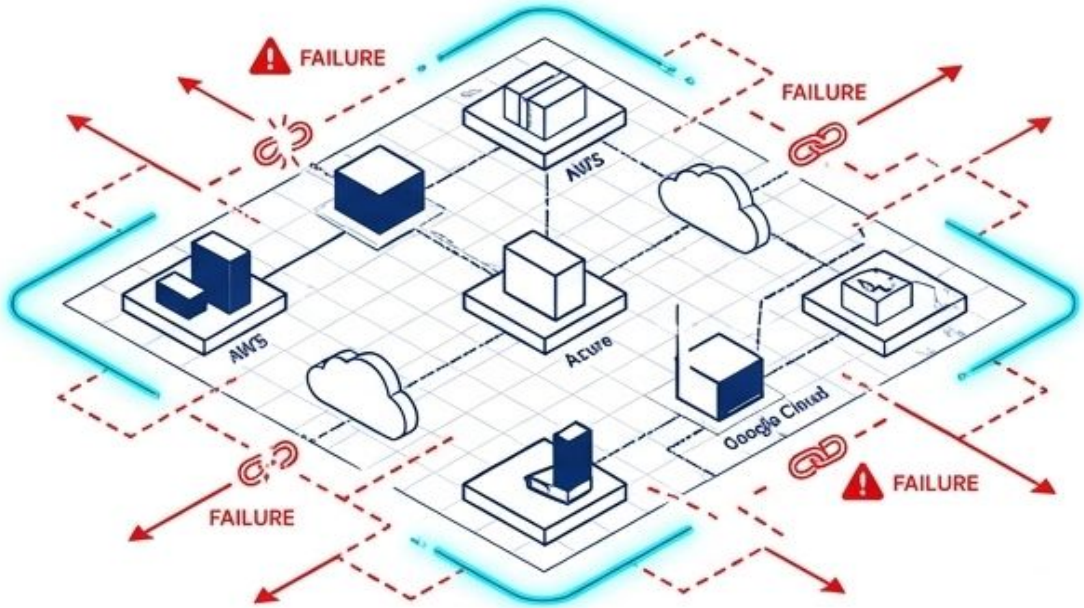
You cannot lift-and-shift a component reliant on a monolithic, highly-coupled processing schedule.

Threat 05: The Quiet Ignorance of Compliance Carve-Outs

ON-PREMISES COMPLIANCE (SECURE)



CLOUD-NATIVE COMPLIANCE (BROKEN)



The Threat

PCI, FedRAMP, or **DORA** compliance achieved on-premises does not **automatically transfer** to the cloud.

The Blast Radius

Teams fail to map old compliance boundaries to new cloud-native controls before migration, leading to **accidental data exposure**.

Sonar Signature

Misconfigurations driven by misunderstood compliance boundaries cause **68-70%** of all cloud security incidents.

Threat 06: The Political Emergency Brake

The Threat:

The migration was approved by the CIO, but the actual application owner (whose bonus depends on stability) was never truly bought in.

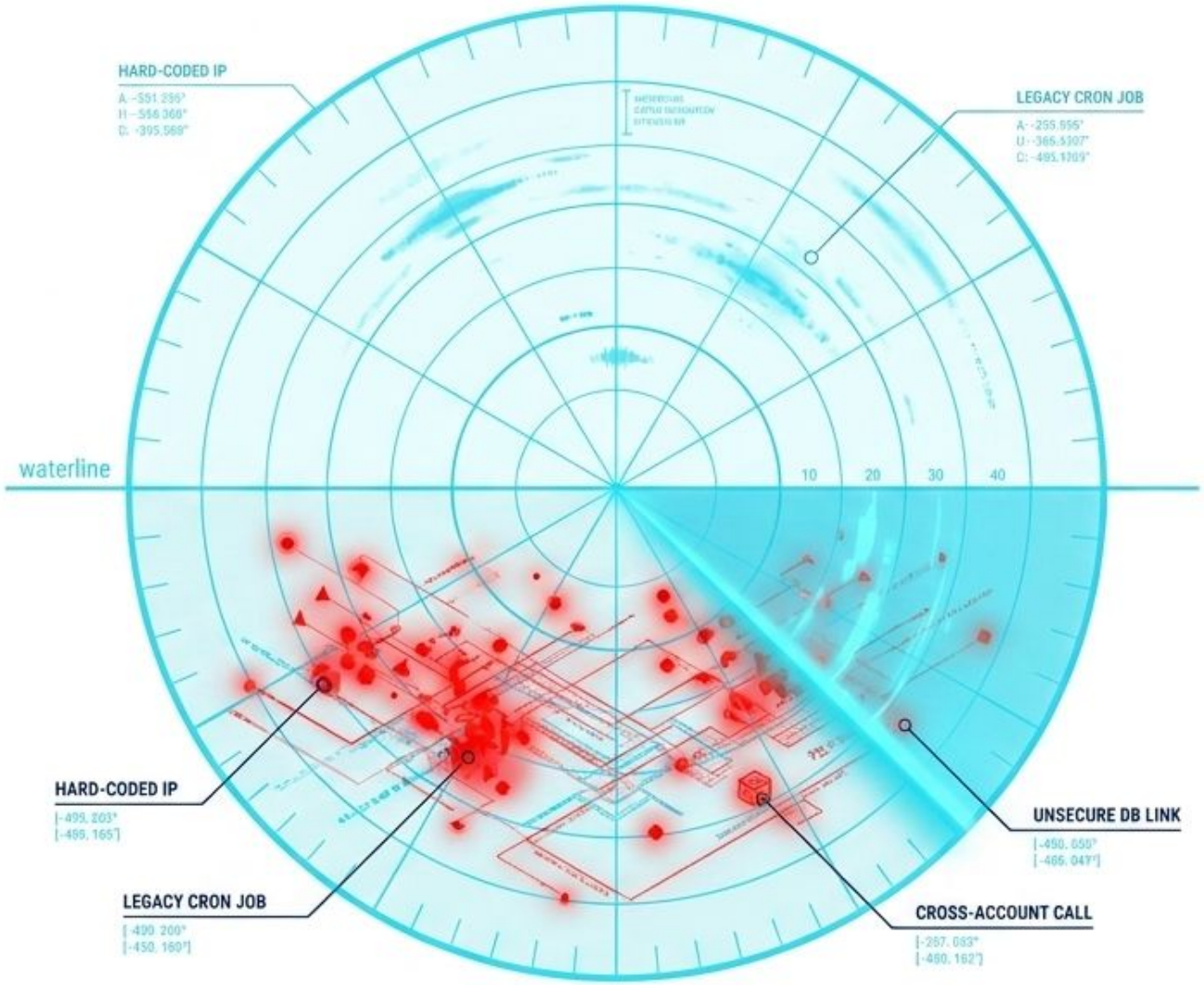
The Blast Radius:

At the first minor hiccup during user acceptance testing, they pull the brake. They use any disruption as proof of unacceptable risk and demand a full rollback.

Sonar Signature:

Look for key stakeholders who were conspicuously absent from the initial architectural planning sessions.

Deploying the Sonar: The Week-Zero Dependency Crawl



The Directive:

Before signing a single cloud contract, you must build a **high-fidelity dependency map. Stop guessing. Start verifying.**

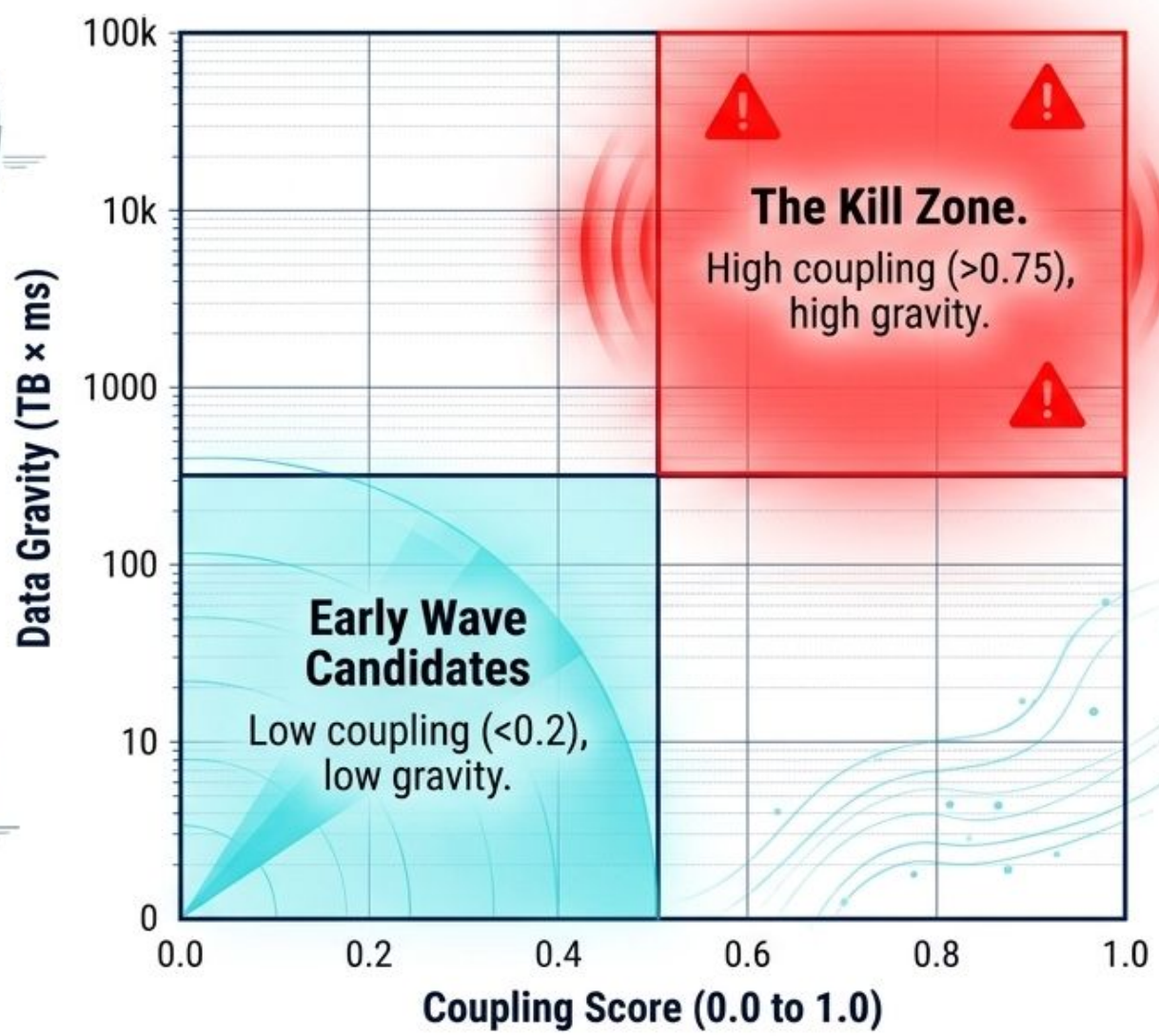
The Output:

Expose **every hard-coded IP address,** forgotten cron job, and **chatty database connection** that will break spectacularly at 3 a.m. during cutover.

The Metric:

The quality of this **Week-Zero** discovery **dictates the success or failure of the entire multi-million dollar program.**

The Predictive Risk Quadrant: Coupling Score vs. Data Gravity



Definitions:

Coupling Score:

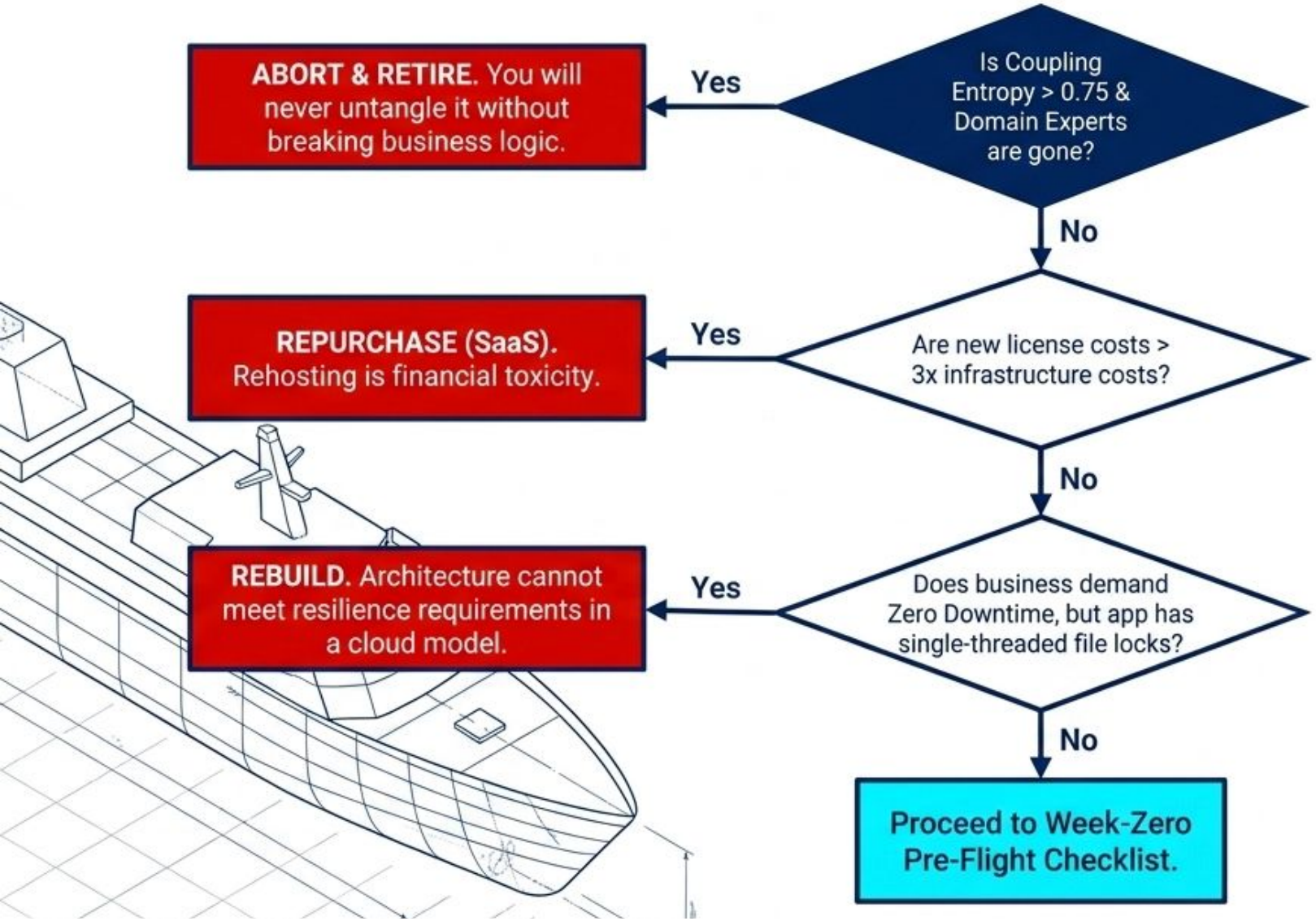
The count of unique inbound/outbound dependencies (API calls, message queues).

A score of 0.9 signals refactoring, not lift-and-shift.

Data Gravity (TB × ms):

The ultimate measure of architectural mass and latency sensitivity.

The Go/No-Go Decision Tree



Draw the Map First, or Don’t Bother Packing.

Great migrations don’t start with blueprints. They start with a complete map of everything that can sink you.

Deploy your Week-Zero Sonar. Expose the dependencies. Command the migration.

Thank you

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